

Handling Rare Mutations in NSCLC Survival Models

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Introduction:

NSCLC remains a leading cause of cancer-related mortality worldwide. Accurate survival prediction is critical, as it allows oncologists to identify high-risk patients who may require aggressive therapies. Genomic sequencing nowadays provides us with information that can better evaluate the cancer progression, which are rely on complex, highly individualized genomic mutational profiles. However, introducing high-dimensional genomic data into survival models faces major statistical challenge. While clinical guidelines heavily emphasize a few driver mutations such as KRAS (Liguori et al., 2025), genomic data inherently has a "long tail" of rare mutations. Traditional Cox Proportional Hazards model struggles with this sparsity that causes overfitting and non-convergence. Thus, researchers employ arbitrary frequency filtering, discarding rare gene mutation information of the cohort, despite their potential value in late-stage mortality estimation.

This study compares methods for incorporating rare mutations into survival analysis of NSCLC cohorts, focusing on the long-tail distribution nature of genomic sequencing data while preserving informative genomic signals. First, this study hypothesizes that rare gene mutations carry important signals for long-term cancer survival prediction despite their low frequency, carrying meaningful information for survival prediction. We further hypothesize that non-linear machine learning will significantly outperform traditional linear models in capturing the complex, multi-gene interactions. By identifying effective strategies to model rare mutations, this study aims to enhance the development of precision risk prediction tools for oncology.

Methods:

Data Acquisition and Study Cohort

Data were retrieved from the MSK cBioPortal data repository, which comprised of both clinical and genomic mutational data. The study cohort was restricted to individuals diagnosed with NSCLC. Clinical characteristics, including age at diagnosis, gender, smoking status, disease stage, and prior medication history, were found statistically significant in previous interim study for the cohort and integrated to the survival models. The primary endpoint for all models was overall survival, defined by left-truncated entry time, exit time, and patient survival status.

Feature Engineering Strategies

Three feature engineering strategies were implemented to the data respectively. The Baseline Frequency Filtering approach used a strict minimum threshold, discarding all gene mutation present in less than 3% of the study cohort. The rarity weighting with TF-IDF method utilized a weighting scheme from natural language processing to assign greater importance to rare mutation genes. In this project, to prevent overfitting anomalies, genes mutated in fewer than three patients were discarded. The remaining genes were weighted by multiplying their binary indicators by inverse document frequency, defined as $\log$ of the total number of patients divided by the mutation frequency within the cohort. The Biological Pathway Aggregation method mapped the genomic data to established biological networks using the KEGG from MSigDB (Kanehisa et al., 2023). The fractional scores were calculated by dividing the number of mutated genes of patient within a specific KEGG pathway by the total number of genes comprising that pathway.

Statistical Modeling and Machine Learning

Two survival modeling methods were applied for each feature engineering strategy. A Cox Proportional Hazards model with LASSO was used to identify highly predictive

feature subsets, with Harrell's C-index calculated via internal k-fold cross-validation during the model fitting process. A post-LASSO unpenalized Cox model was then constructed using selected features and clinical covariates. Additionally, to capture complex, non-linear gene interactions, Random Survival Forests (RSF) were applied using an 80/20 train-test split. For this approach, Harrell's C-index was explicitly calculated as an out-of-sample metric using the strictly held-out 20% test set. Overall performance for both modeling frameworks was further assessed utilizing Time-Dependent AUC evaluated at the 80th percentile clinical follow-up time point.

Results:

Feature Engineering Outcomes

The analytical cohort comprised 7,802 patients diagnosed with NSCLC, with an initial genomic profile spanning 517 distinct genes. Application of the Baseline Frequency Filtering strategy eliminated 454 genes, retaining only 63 gene mutations for model training. Conversely, the TF-IDF weighting strategy preserved a significantly larger proportion of the genomic information, retaining 509 genes after dropping those occurring in fewer than three patients. The Biological Pathway Aggregation mapped gene mutations to 112 pathways contained at least one mutation within the study cohort.

Model Performance Evaluation

For the Baseline Frequency model, well-known driver mutations such as TP53, KEAP1, EGFR, STK11, and PTEN were identified statistically significant. Its cross-validated mean C-index was 0.713. When evaluated at the 80th percentile of follow-up time (66.4 months), the baseline Cox model produced a time-dependent AUC of 0.730. The TF-IDF Cox model demonstrated comparable performance of 0.718 C-index and AUC of 0.740. The Biological Pathway Aggregation Cox model produced a

time-dependent AUC of approximately 0.729. In summary, Lasso Cox models demonstrated stable, consistent concordance for all three feature engineering strategies.

Random Survival Forests were trained using an 80/20 train-test split. The Baseline RSF achieved an out-of-sample C-index of 0.694 but showed better performance in late-stage time-dependent evaluation, producing an AUC of 0.815 at the 80th percentile time horizon. The Biological Pathway Aggregation RSF also demonstrated late-stage discriminatory power, presenting C-index of only 0.674 while achieving an out-of-sample AUC of approximately 0.802. Similar robust non-linear performance was observed in the TF-IDF RSF with AUC of 0.812, with a slightly higher c-index of 0.689. These results validating that machine learning ensembles can effectively process high-dimensional rare mutations and complex interactions between genes.

Discussion:

By evaluating distinct feature engineering strategies across both linear and non-linear modeling approaches, our results suggest that including rare mutations still leads to stable predictive performance, but the choice of survival model plays a major role in how accurately patient survival status is estimated over time.

The first hypothesis suggested that rare gene mutations may contain important information for predicting long-term survival. By using rarity weighting and Biological Pathway Aggregation methods, we were able to keep gene information up to 509 genes and 112 pathways compared with just 63 genes kept by frequency filtering. Importantly, adding this larger and more complex set of features did not harm model stability or convergence. The Lasso Cox models performed consistently across all

approaches, with rarity weighting method performed slightly better. This suggests that rare mutations do provide useful predictive signals. However, the improvement was small, probably since driver mutations such as TP53, KEAP1, and EGFR still dominate the model. As a result, the cox model has limited ability to fully take advantage of the additional rare mutation data.

The evaluation of our second hypothesis, which hypothesized that non-linear machine learning approach would significantly outperform traditional linear models, revealed a clear divergence between global concordance and time-specific discrimination. The Lasso Cox models showed superior global C-indices, suggesting a strong capability in ranking patient risk across the entire study period. Random Survival Forests demonstrated lower performance in this perspective but obtained substantially higher time dependent AUCs at the 80th percentile follow-up horizon. The ensemble approach validated its capability to capture the complex, multi-gene interactions. This performance gap highlights that non-linear machine learning is better at modeling late-stage mortality risks that cox proportional hazards models fail to recognize.

Several limitations should be considered when interpreting these findings. First, this study relied on a single retrospective cohort with single cancer type, which may limit generalizability to a broader, diverse patient population. Second, the threshold choices and pathway mappings using KEGG introduced were arbitrary, which could influence results. Additionally, While RSF demonstrated improved time-specific discrimination, its interpretability remains limited compared to more transparent coefficient estimation provided by Cox-based models.

Reference:

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Table 1: Comparative Performance of Survival Models Across Genetic Feature Strategies

Feature Engineering Strategy	Lasso-Cox C-Index	RSF C-Index	Lasso-Cox AUC	Random Forest AUC
Frequency Filtering	0.713	0.694	0.730	0.815
Rarity Weighting	0.718	0.689	0.735	0.812
Pathway Aggregation	0.708	0.674	0.729	0.802

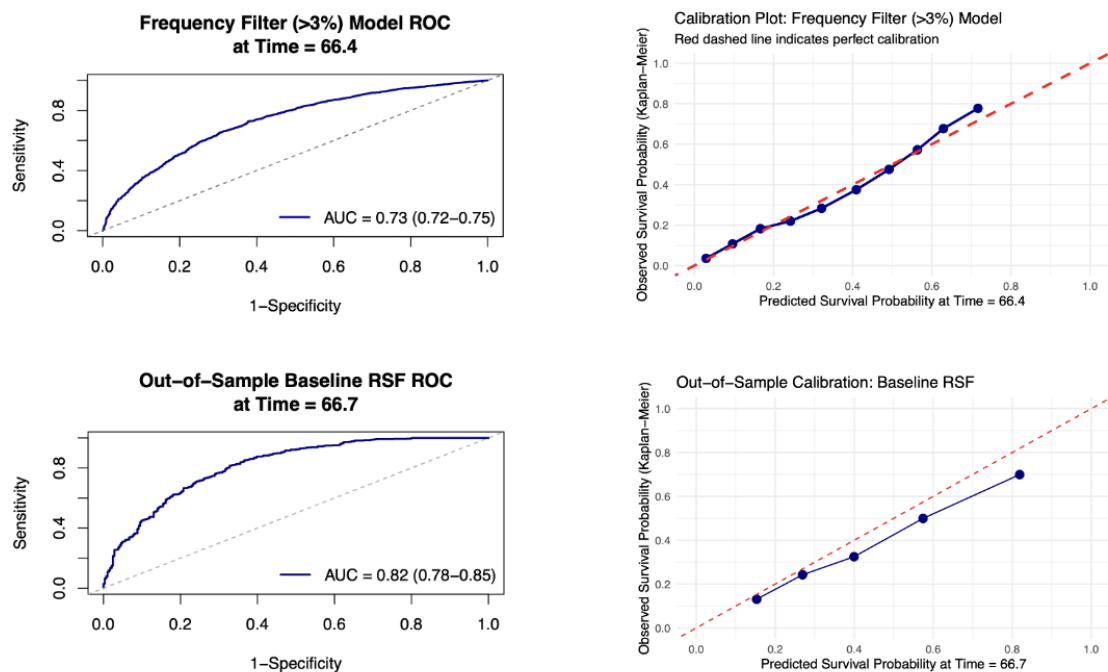


Figure 1: Predictive Performance and Calibration Analysis of Survival Models with Frequency Filtering Strategy

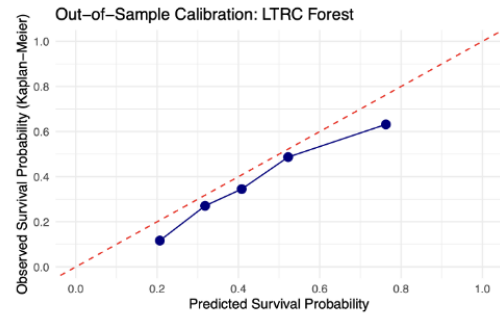
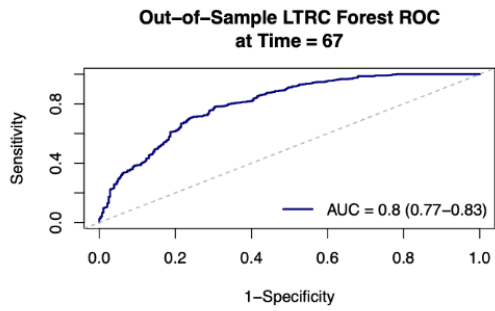
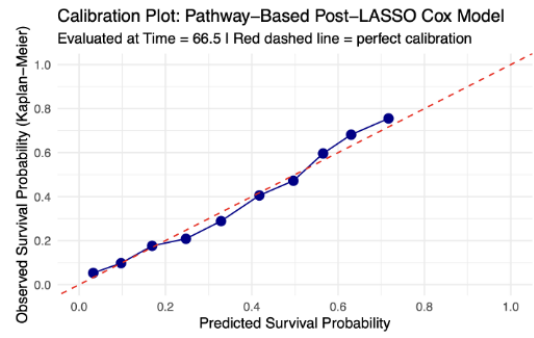
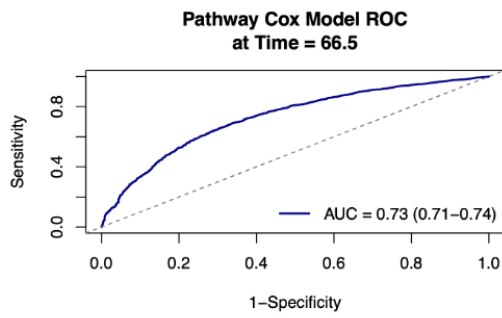


Figure 2: Predictive Performance and Calibration Analysis of Survival Models with Pathway Aggregation Strategy

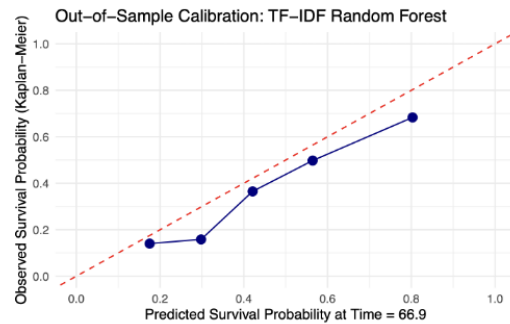
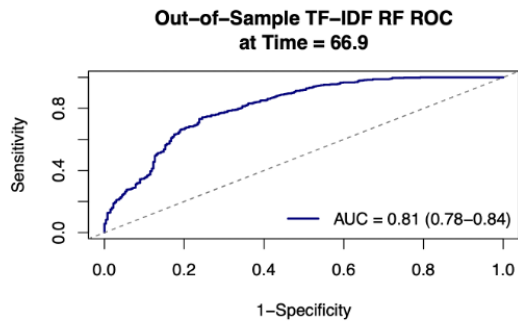
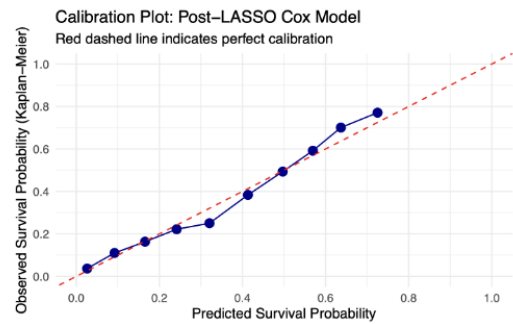
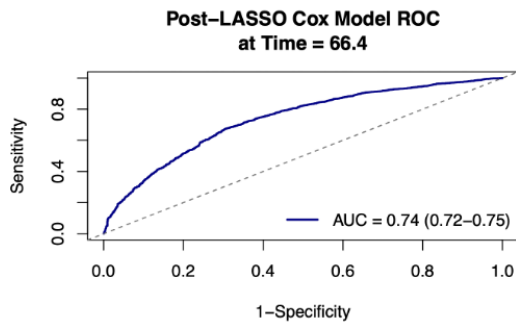


Figure 3: Predictive Performance and Calibration Analysis of Survival Models with Rarity Weighting Strategy